

Math 221 Homework 9

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Question 5.7. Let q be a unit vector and d any vector with $q^T d = 0$. Show that

$$\|(q + d)q^T - I\|_2 = \|q + d\|_2.$$

Proof. If $d = 0$ then $(q + d)q^T - I = qq^T - I$ has eigenvalues 0 and -1 , so $\|(q + d)q^T - I\|_2 = 1 = \|q\|_2 = \|q + d\|_2$ and we are done. Assume $d \neq 0$.

Set $\alpha = \|d\|_2$ and $u_2 = d/\alpha$. Complete $\{q, u_2\}$ to an orthonormal basis $\{q, u_2, u_3, \dots, u_n\}$ of $\mathbb{R}^n$. Consider

$$A := (q + d)q^T - I.$$

We compute A on this basis. First,

$$Aq = (q + d)q^T q - q = (q + d) - q = d = \alpha u_2,$$

and since $q^T u_2 = q^T d/\alpha = 0$,

$$Au_2 = (q + d)q^T u_2 - u_2 = -u_2.$$

For $j \geq 3$ we have $q^T u_j = 0$, so $Au_j = -u_j$.

Thus, in the orthonormal basis $\{q, u_2, u_3, \dots, u_n\}$, the matrix of A is block diagonal with a 2×2 block

$$B = \begin{pmatrix} 0 & 0 \\ \alpha & -1 \end{pmatrix}$$

and a block $-I_{n-2}$ on the remaining coordinates. The singular values of A are therefore those of B together with 1 (with multiplicity $n - 2$), so

$$\|A\|_2 = \max\{\|B\|_2, 1\}.$$

Now

$$B^T B = \begin{pmatrix} 0 & \alpha \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ \alpha & -1 \end{pmatrix} = \begin{pmatrix} \alpha^2 & -\alpha \\ -\alpha & 1 \end{pmatrix}.$$

The characteristic polynomial is

$$\det(B^T B - \lambda I) = (\alpha^2 - \lambda)(1 - \lambda) - \alpha^2 = \lambda(\lambda - (1 + \alpha^2)),$$

so the eigenvalues of $B^T B$ are 0 and $1 + \alpha^2$. Hence the singular values of B are 0 and $\sqrt{1 + \alpha^2}$, and

$$\|B\|_2 = \sqrt{1 + \alpha^2} \geq 1.$$

Therefore $\|A\|_2 = \sqrt{1 + \alpha^2}$.

Finally, since q is unit and orthogonal to d ,

$$\|q + d\|_2^2 = \|q\|_2^2 + \|d\|_2^2 = 1 + \alpha^2,$$

so $\|q + d\|_2 = \sqrt{1 + \alpha^2} = \|A\|_2$. This proves

$$\|(q + d)q^T - I\|_2 = \|q + d\|_2.$$

□

Question 5.9. Prove bound (5.6) from Theorem 5.5. Let A be symmetric with orthonormal eigenvectors $q_1, \dots, q_n$ and $Aq_j = \alpha_j q_j$. Let x be a unit vector, let

$$\rho = \rho(x, A), \quad r = Ax - \rho x,$$

and let θ be the angle between x and q_i . Show that

$$\sin \theta \leq \frac{\|r\|_2}{\text{gap}'}, \quad \text{gap}' := \min_{j \neq i} |\alpha_j - \rho|.$$

Proof. Since $\{q_j\}$ is an orthonormal basis, write

$$x = \sum_{j=1}^n b_j q_j, \quad \sum_{j=1}^n b_j^2 = 1.$$

Using $Ax - \rho x = r$ and $Aq_j = \alpha_j q_j$ we obtain

$$(A - \rho I) \sum_{j=1}^n b_j q_j = r, \quad b_i Aq_i = b_i \alpha_i q_i.$$

Subtracting gives

$$(A - \rho I) \sum_{j \neq i} b_j q_j = r + (\rho - \alpha_i) b_i q_i.$$

The left-hand side has no q_i component, so the q_i component on the right must vanish. In other words, the q_i component of r is cancelled by $(\rho - \alpha_i) b_i q_i$, and hence

$$\|r + (\rho - \alpha_i) b_i q_i\|_2 \leq \|r\|_2.$$

Write

$$r + (\rho - \alpha_i) b_i q_i = \sum_{j \neq i} c_j q_j.$$

Then

$$(A - \rho I) \sum_{j \neq i} b_j q_j = \sum_{j \neq i} b_j (\alpha_j - \rho) q_j = \sum_{j \neq i} c_j q_j,$$

so for each $j \neq i$,

$$b_j = \frac{c_j}{\alpha_j - \rho},$$

and we assume $\text{gap}' > 0$ so that the denominators are nonzero.

Now

$$\sin \theta = \left\| \sum_{j \neq i} b_j q_j \right\|_2 = \left(\sum_{j \neq i} |b_j|^2 \right)^{1/2} = \left(\sum_{j \neq i} \frac{|c_j|^2}{|\alpha_j - \rho|^2} \right)^{1/2} \leq \frac{1}{\text{gap}'} \left(\sum_{j \neq i} |c_j|^2 \right)^{1/2}.$$

By the inequality above,

$$\left(\sum_{j \neq i} |c_j|^2 \right)^{1/2} = \|r + (\rho - \alpha_i) b_i q_i\|_2 \leq \|r\|_2,$$

and therefore

$$\sin \theta \leq \frac{\|r\|_2}{\text{gap}}.$$

This is exactly bound (5.6).

□

Question 5.13. Let A be a real symmetric matrix. Run shifted QR iteration (Algorithm 4.5) on A with Rayleigh quotient shift, i.e. at step k use $\sigma_k = (A_k)_{nn}$, where A_k is the current iterate. This produces a sequence of shifts $\{\sigma_k\}$. Also run Rayleigh quotient iteration (Algorithm 5.1) on A , starting from $x_0 = e_n = [0, \dots, 0, 1]^T$, which produces a sequence of Rayleigh quotients $\{\rho_k\}$. Show that $\sigma_k = \rho_k$ for all k .

Proof. We compare the two algorithms inductively, coupling them so that they use the same orthogonal factors.

First note that

$$\sigma_1 = (A_1)_{nn} = e_n^T A e_n,$$

while Rayleigh quotient iteration starts from $x_0 = e_n$, so

$$\rho_1 = x_0^T A x_0 = e_n^T A e_n = \sigma_1.$$

Thus $\sigma_1 = \rho_1$.

Assume now that after k steps the shifted QR iteration has produced a symmetric matrix A_k and an orthogonal matrix P_{k-1} such that

$$A_k = P_{k-1}^T A P_{k-1},$$

and that Rayleigh quotient iteration has a current vector

$$x_{k-1} = P_{k-1} e_n$$

so that

$$\rho_k = x_{k-1}^T A x_{k-1} = e_n^T A_k e_n = (A_k)_{nn}.$$

Thus the shift used at step k of the QR iteration can be written as

$$\sigma_k = (A_k)_{nn} = \rho_k.$$

Now perform step k of both algorithms using the common shift ρ_k and a common QR factorization. Factor

$$A_k - \rho_k I = Q_k R_k,$$

with Q_k orthogonal and R_k upper triangular, and set

$$P_k = P_{k-1} Q_k, \quad A_{k+1} = R_k Q_k + \rho_k I.$$

As usual in the QR algorithm, we have

$$A_{k+1} = Q_k^T A_k Q_k = P_k^T A P_k,$$

so the new iterate is again orthogonally similar to A .

On the Rayleigh quotient side, the next vector is obtained by inverse iteration with the same shift:

$$(A - \rho_k I) y_k = x_{k-1}, \quad x_k = \frac{y_k}{\|y_k\|_2}.$$

Using $A_k = P_{k-1}^T A P_{k-1}$ and $x_{k-1} = P_{k-1} e_n$, this system is equivalent to

$$(A_k - \rho_k I) \hat{y}_k = e_n, \quad \hat{y}_k = P_{k-1}^T y_k,$$

so $\hat{y}_k = R_k^{-1} Q_k^T e_n$ up to a nonzero scalar. Thus the normalized solution can be written as

$$x_k = P_{k-1} \frac{\hat{y}_k}{\|\hat{y}_k\|_2} = P_{k-1} Q_k z_k = P_k z_k$$

for some unit vector z_k . In particular, we may choose our orthogonal factor Q_k so that $z_k = e_n$ (this just fixes the last column of Q_k), and then

$$x_k = P_k e_n.$$

With this choice, the Rayleigh quotient at the next step is

$$\rho_{k+1} = x_k^T A x_k = e_n^T P_k^T A P_k e_n = e_n^T A_{k+1} e_n = (A_{k+1})_{nn} = \sigma_{k+1}.$$

Hence $\sigma_{k+1} = \rho_{k+1}$, completing the induction. Therefore $\sigma_k = \rho_k$ for all k , as claimed. $\square$

Question 5.15. Suppose

$$t(n) = 2t\left(\frac{n}{2}\right) + cn^3 + O(n^2).$$

Show that

$$t(n) \approx \frac{4}{3}cn^3,$$

which justifies the complexity estimate for the divide-and-conquer algorithm.

Proof. Assume $n = 2^k$ and unroll the recurrence for $k = \log_2 n$ levels. At level j of the recursion tree ($j = 0, 1, \dots, k-1$), there are 2^j subproblems, each of size $n/2^j$, so the cubic work at that level is

$$2^j \cdot c \left(\frac{n}{2^j}\right)^3 = cn^3 2^{j-3j} = cn^3 4^{-j}.$$

Summing over all levels gives the total cubic cost

$$cn^3 \sum_{j=0}^{k-1} 4^{-j} = cn^3 \cdot \frac{1 - 4^{-k}}{1 - \frac{1}{4}} = \frac{4}{3}c(1 - 4^{-k})n^3.$$

Since $4^{-k} = 2^{-2k} = n^{-2}$, this is

$$\frac{4}{3}c(1 - n^{-2})n^3.$$

The contribution from the leaves is $2^k t(1) = nt(1) = O(n)$, and all accumulated lower-order terms from the $O(n^2)$ in the recurrence sum to $O(n^2)$. Hence

$$t(n) = \frac{4}{3}c(1 - n^{-2})n^3 + O(n^2) = \frac{4}{3}cn^3 + O(n^2),$$

so asymptotically $t(n) \approx \frac{4}{3}cn^3$. □

Question 5.16. Let $A = D + \rho uu^T$, where $D = \text{diag}(d_1, \dots, d_n)$ and $u = [u_1, \dots, u_n]^T$. Show that if $d_i = d_{i+1}$ or $u_i = 0$, then d_i is an eigenvalue of A . If $u_i = 0$, show that the corresponding eigenvector is e_i , the i th column of the identity. Derive a similarly simple expression when $d_i = d_{i+1}$. This explains how to handle deflation in the divide-and-conquer algorithm.

Proof. First suppose $u_i = 0$. Then the i th column of uu^T is zero, so

$$Ae_i = (D + \rho uu^T)e_i = De_i + \rho u(u^T e_i) = d_i e_i,$$

and hence d_i is an eigenvalue with eigenvector e_i .

Now suppose $d_i = d_{i+1}$, and assume $u_i \neq 0$ (otherwise we are back in the previous case; if $u_{i+1} = 0$ instead, we can swap the roles of i and $i+1$). Consider

$$v := e_{i+1} - \frac{u_{i+1}}{u_i} e_i.$$

Since D is diagonal and $d_i = d_{i+1}$,

$$Dv = De_{i+1} - \frac{u_{i+1}}{u_i} De_i = d_{i+1}e_{i+1} - \frac{u_{i+1}}{u_i} d_i e_i = d_i v.$$

For the rank-one term, note that

$$u^T v = u_{i+1} - \frac{u_{i+1}}{u_i} u_i = 0,$$

so $uu^T v = u(u^T v) = 0$. Therefore

$$Av = (D + \rho uu^T)v = Dv + \rho uu^T v = d_i v,$$

showing that v is an eigenvector of A with eigenvalue d_i .

Thus, in either case ($u_i = 0$ or $d_i = d_{i+1}$) we obtain d_i as an eigenvalue with an explicitly known eigenvector, which can then be used for deflation in the divide-and-conquer scheme. $\square$

Proof. Let $A \in \mathbb{R}^{m \times n}$ with $m \geq n$, and take a reduced SVD

$$A = U\Sigma V^T,$$

where $U \in \mathbb{R}^{m \times n}$ has orthonormal columns, $V \in \mathbb{R}^{n \times n}$ is orthogonal, and $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_n)$ with $\sigma_i \geq 0$.

Define

$$Q := UV^T, \quad P := V\Sigma V^T.$$

Then

$$Q^T Q = VU^T UV^T = VIV^T = I,$$

so Q has orthonormal columns. Also P is symmetric and positive semidefinite, since

$$P = V\Sigma V^T = (\Sigma^{1/2} V^T)^T (\Sigma^{1/2} V^T).$$

Finally,

$$QP = UV^T V \Sigma V^T = U \Sigma V^T = A,$$

so $A = QP$ is a polar decomposition.

For uniqueness, assume A has full column rank and suppose

$$A = QP = \hat{Q}\hat{P}$$

are two polar decompositions with $Q, \hat{Q}$ having orthonormal columns and $P, \hat{P}$ positive definite. Then

$$A^T A = P^2 = \hat{P}^2.$$

The positive definite square root of $A^T A$ is unique, so $P = \hat{P}$, and hence

$$Q = AP^{-1} = A\hat{P}^{-1} = \hat{Q}.$$

Thus, when A has full column rank (in particular, when it is square and nonsingular), the polar decomposition is unique. $\square$

Question 5.23. Let A be an $n \times n$ symmetric positive definite matrix with Cholesky factorization $A = LL^T$, and let $\hat{L}$ be the Cholesky factor computed in floating-point arithmetic. We want to bound the relative error in the (squared) singular values of $\hat{L}$ as approximations to the eigenvalues of A .

Proof. First scale A so that the diagonal is 1. Put

$$D := \text{diag}(a_{11}^{1/2}, \dots, a_{nn}^{1/2}), \quad \bar{A} := D^{-1}AD^{-1},$$

so $A = D\bar{A}D$ and

$$\bar{a}_{ii} = e_i^T D^{-1}AD^{-1}e_i = \frac{a_{ii}}{a_{ii}} = 1 \quad \text{for all } i.$$

Write $L = DX$. Then

$$A = LL^T = DXX^TD = D\bar{A}D \implies \bar{A} = XX^T.$$

The eigenvalues of $\bar{A}$ are therefore the squares of the singular values of X , so

$$\kappa(\bar{A}) = \frac{\lambda_{\max}(\bar{A})}{\lambda_{\min}(\bar{A})} = \frac{\sigma_{\max}(X)^2}{\sigma_{\min}(X)^2} = \kappa(X)^2.$$

By backward stability of Cholesky (bound (2.16)), the computed factor $\hat{L}$ satisfies

$$A + \delta A = \hat{L}\hat{L}^T,$$

with

$$\|\bar{A}^{-1/2} \delta \bar{A} \bar{A}^{-1/2}\|_2 = O(\varepsilon) \kappa(\bar{A}), \quad \delta \bar{A} := D^{-1} \delta A D^{-1}.$$

Equivalently, for $B := L^T L$ and $\hat{B} := \hat{L}^T \hat{L}$, we have

$$\hat{B} = B + E, \quad \|B^{-1/2} E B^{-1/2}\|_2 = O(\varepsilon) \kappa(\bar{A}).$$

Theorem 5.6 (applied to B and $\hat{B}$) then implies that there exists an invertible matrix Y such that

$$\hat{L}^T \hat{L} = \hat{B} = Y^T B Y = Y^T L^T L Y,$$

with

$$\|Y^T Y - I\|_2 \leq O(\varepsilon) \kappa(\bar{A}),$$

and that the eigenvalues $\lambda_i(\hat{B})$ and $\lambda_i(B)$ satisfy the relative bound

$$\frac{|\lambda_i(\hat{B}) - \lambda_i(B)|}{\lambda_i(B)} \leq O(\varepsilon) \kappa(\bar{A}) \quad \text{for all } i.$$

Next, note that $L^T L$ and LL^T are similar:

$$LL^T = L(L^T L)L^{-1},$$

so they have the same eigenvalues; similarly $\hat{L}\hat{L}^T$ and $\hat{L}^T \hat{L}$ are similar and share eigenvalues. Hence the same relative eigenvalue bound holds for the pair $\hat{L}\hat{L}^T$ and LL^T .

Finally, the eigenvalues of $LL^T = A$ are the squares of the singular values of L , and the eigenvalues of $\hat{L}^T \hat{L}$ (or $\hat{L}\hat{L}^T$) are the squares of the singular values of $\hat{L}$. Thus the squared singular values of $\hat{L}$ approximate the eigenvalues of A with relative error

$$O(\varepsilon) \kappa(\bar{A}) = O(\varepsilon) \kappa(X)^2 \sim O(\varepsilon) \kappa(L)^2,$$

as claimed. □